DIGITAL MARKETING HEROES
digitalheroes.co.in

SOFTWARE REPORT

SOFTWARE DOCUMENTATION

AirTrainr Web Application

Complete Technical Documentation & Feature Specification

PREPARED BY
Digital Marketing Heroes

DATE
April 17, 2026

VERSION
2.0

Table of Contents

01 Executive Summary & Project Overview 03
- 1.1 Platform Vision & Goals 03
- 1.2 Technology Stack 03
- 1.3 Key Performance Indicators 03

02 System Architecture & Database Design 04
- 2.1 Monorepo Structure 04
- 2.2 Database Schema (15+ tables) 04
- 2.3 Third-Party Integrations 05

03 User Roles & Feature Specifications 06
- 3.1 Athlete Features (12 modules) 06
- 3.2 Trainer Features (14 modules) 07
- 3.3 Admin Panel (10 modules) 08

04 Business Workflows & User Journeys 09

05 Payment Architecture & Financial Model 10

06 Location Intelligence & Internationalization 11

07 Security, Compliance & Data Protection 12

08 Testing, QA & Deployment 13

01

SECTION ONE

Executive Summary

Platform vision, technology decisions, and key metrics

1.1 Platform Vision & Goals

AirTrainr is an intelligence-driven sports training marketplace designed to bridge the gap between athletes seeking professional coaching and certified trainers looking to grow their business. Operating across the United States and Canada, the platform supports 20+ sports disciplines and provides end-to-end management for the entire training lifecycle — from discovery and booking to payment and review.

The platform addresses three core problems: (1) athletes struggle to find qualified, local trainers for their specific sport and skill level; (2) independent trainers lack tools to manage their business professionally; and (3) there is no trusted payment and escrow system for training sessions in the sports coaching market.

> Mission Statement: Democratize access to elite sports training by connecting athletes of all levels with verified, professional trainers in their area — while providing trainers with the tools to build and scale their coaching business.

1.2 TECHNOLOGY STACK

Component	Technology	Rationale
Frontend Framework	Next.js 16 + React 19	Server-side rendering, API routes, edge functions
Database	Supabase (PostgreSQL 15 + PostGIS)	Real-time subscriptions, RLS security, geospatial queries
Payments	Stripe Connect Express + Checkout	Marketplace model, trainer payouts, escrow
Media Storage	Cloudinary	Image/video optimization, CDN delivery, auto-format

Location Services	Google Places API	City autocomplete, geocoding, address validation
Email	Nodemailer + GoDaddy SMTP	Transactional emails, receipts, contact notifications
Real-time	Socket.io + Supabase Realtime	Live messaging, notification delivery
Hosting	Vercel	Edge network, automatic deployments, serverless functions
Testing	Playwright	47 E2E tests across all features

1.3 KEY PERFORMANCE INDICATORS

SPORTS COVERAGE

20+ sports: Hockey, Baseball, Basketball, Football, Soccer, Tennis, Golf, Swimming, Boxing, Lacrosse, Wrestling, Martial Arts, Gymnastics, Track & Field, Volleyball, Skiing, Snowboarding, Figure Skating, Softball, Rugby

MARKET REACH

United States & Canada with admin-configurable country expansion. Location demand tracking captures interest from 15+ potential markets including UK, Australia, India, Germany, France.

REVENUE MODEL

3% platform fee (admin-configurable) charged to athlete on top of trainer rate. Trainer receives 100% of their session price. Subscription: $25/mo or $250/yr for trainers.

FOUNDING 50 PROGRAM

First 50 trainers receive 6 months free Pro access, permanent gold badge in search results, and guaranteed top placement in all search queries across the platform.

02

SECTION TWO

System Architecture

Infrastructure design, database schema, and integration map

2.1 Monorepo Structure

Directory	Technology	Description
apps/web/	Next.js 16	Main web application — 50+ pages, API routes, middleware
services/api/	Express.js	Backend API — booking logic, email, cron jobs, Socket.io
packages/shared/	TypeScript	Shared types, enums, validation utilities across all packages
packages/config/	Configuration	Shared environment and build configuration

The web application uses Next.js App Router with server components, API routes for Stripe/Places proxying, and Turbopack for development. All pages use the shared Supabase client for real-time data.

2.2 DATABASE SCHEMA

Table	Rows	Key Columns	Purpose
users	*	id, email, role, first_name, avatar_url	All platform users (athlete/trainer/admin)
athlete_profiles	*	sports[], skill_level, city, zip_code, travel_radius_miles	Athlete-specific profile data
trainer_profiles	*	bio, hourly_rate, sports[], camp_offerings (JSON), stripe_account_id	Trainer business profile + Stripe Connect
bookings	*	athlete_id, trainer_id, sport, scheduled_at, status, price	Training session bookings with lifecycle
training_offers	*	trainer_id, athlete_id, proposed_dates (JSON), price, status	Trainer-to-athlete session/camp offers
payment_transactions	*	booking_id, amount, platform_fee, trainer_payout, hold_until	Escrow payment records
availability_slots	*	trainer_id, day_of_week, start_time, end_time	Per-slot availability entries
availability_recurring	*	trainer_id, day_of_week, start_time, end_time, is_active	Weekly recurring schedule
notifications	*	user_id, type, title, body, data (JSON), read	Push notification records
messages	*	booking_id, sender_id, content, read_at	In-app messaging per booking
reviews	*	booking_id, rating, comment, reviewer_id	Post-session ratings & feedback
sub_accounts	*	parent_user_id, profile_data (JSON), is_active	Family member athlete accounts (max 6)
contact_messages	*	email, subject, message, user_id	Contact form submissions
location_leads	*	searched_city, searched_country	Demand from unavailable regions
receipts	*	booking_id, recipient_role, session_fee, platform_fee	Email receipt records
platform_settings	1	platform_fee_percentage, allowed_countries[], maintenance_mode	Global configuration

2.3 Third-Party Integrations

Service	Integration	Endpoints / Usage
Stripe Connect	Express accounts for trainer payouts	/api/stripe/connect, /api/stripe/connect-status
Stripe Checkout	Athlete session payments	/api/stripe/create-booking-payment, /api/stripe/create-offer-payment
Stripe Webhooks	Payment confirmation + booking creation	/api/stripe/webhook (checkout.session.completed, invoice.*)
Google Places	City autocomplete + geocoding	/api/places/autocomplete, /api/places/details (server proxy)
Cloudinary	Image upload + CDN	Client-side upload with preset "airtrainr", folders: avatars, sports, profiles
Supabase Realtime	Live notifications + message sync	postgres_changes subscription on messages, notifications tables
GoDaddy SMTP	Transactional emails	contact@airtrainr.com via smtpout.secureserver.net:465

API ROUTES SUMMARY (NEXT.JS)

Route	Method	Purpose
/api/stripe/connect	POST/GET	Create Stripe account + check status
/api/stripe/create-offer-payment	POST	Checkout session for offer acceptance
/api/stripe/create-booking-payment	POST	Checkout session for direct booking
/api/stripe/verify-subscription	POST	Fallback subscription activation
/api/stripe/webhook	POST	Stripe event handler
/api/places/autocomplete	GET	Google Places city search proxy
/api/places/details	GET	Place details with address components
/api/contact	POST	Contact form + email notification
/api/availability	POST	Save recurring weekly schedule
/api/admin/*	Various	Settings, payouts, trainer management

03

SECTION THREE

Feature Specifications

Detailed capabilities for each user role

3.1 Athlete Features

Module	Features
Registration	Email/password (12+ chars), Google OAuth, role selection, sport selection, city (Google Places)
Profile	Avatar upload (Cloudinary + pencil badge), first/last name, phone, DOB, sex, bio, address, city (autocomplete), state, ZIP (cross-validated with country), travel radius (auto km/mi for Canada)
Sports	Multi-select from 20+ sports, proper capitalization (formatSportName utility)
Search & Discovery	Sport filter, sort (relevance/rating/price/distance), location filter, Founding 50 priority, map view with pins, verified badges
Trainer Calendar	Green-highlighted available dates (60-day window), slot selection by duration, real-time booking conflict check
Booking	Select slot → Stripe Checkout → booking confirmed. Status lifecycle: pending → confirmed → completed
Offer Response	Receive trainer offers → Accept (Stripe payment) or Decline. Camp offers show auto-filled schedule.
Messaging	Real-time chat per booking, read receipts, unread badges, message history
Sub-Accounts	Up to 6 family members, multi-sport (up to 3 per account), backwards-compatible legacy support
Payments	Payment history, escrow status, total spent, upcoming sessions
Notifications	Real-time push: booking updates, offers, session reminders (24h + 1h), reviews

Contact	Form with subject dropdown, 500-char message, email to admin + Supabase storage

3.2 Trainer Features

Module	Features
Profile Setup	Photo upload (Cloudinary → avatar_url sync), bio, headline, certifications, hourly rate, package rate, years experience, sports selection
Location	City (Google Places autocomplete), state (auto-filled), ZIP (cross-validated), travel radius (km for CA, mi for US), country auto-detect
Availability	**Recurring Weekly:** toggle per day with start/end time, saved via API route. **Per-Slot:** individual 30/45/60 min slots with conflict detection.
Camps	Name, total price, hours/day. **Schedule:** non-consecutive days with individual start time per day (end time auto-calculated). Location (autocomplete). Max spots with progress bar (green/amber/red). Race condition protection on accept (3x retry loop).
Training Offers	Browse athletes, select sport, add multiple date/time entries. Camp select → auto-fills schedule (read-only). Timezone detection + display. Intro discount toggle (10-50%).
Subscription	7-day free trial, $25/mo or $250/yr. Verification pending banner. Founding 50 application. Stripe Checkout for payment.
Payment Settings	Stripe Connect Express onboarding (3 states: not connected / partial / active). Bank last4 display. Dashboard link. Auth-protected APIs.
Earnings	Total earned, net earnings, in escrow. Upcoming sessions with hold dates. Completed session history. CSV export (ISO dates, no platform fee column).
Sessions	Upcoming + completed bookings. Status management (confirm/reject/complete). Reschedule support.
Reviews	View ratings & comments from athletes. Performance verification (3+ sessions, 95%+ completion, 0 disputes).
Messaging	Same as athlete — real-time per-booking chat
Verification	Document upload (PDF), admin review workflow, status badges (pending/verified/rejected)

3.3 Admin Panel

Module	Capabilities
Dashboard	Platform stats, revenue overview, user growth metrics, quick actions
Trainer Management	Table with search/filter, verify/reject trainers, approve profile images, view documents, map view with click-to-profile (z-index 9999 for modal above map)
Athlete Management	Table with full location display (city, state, country, ZIP), detail modal, horizontal scroll for wide tables
Payment Management	Escrow overview, release single/batch payouts, CSV export with properly quoted dates (YYYY-MM-DD), pending/ready/released tracking
Sports Management	Add/edit sports, Cloudinary image upload per sport, active/inactive toggle
Subscriptions	Manage trainer subscriptions, Founding 50 approval/rejection, trial tracking
Disputes	Review booking disputes, resolve (refund athlete / payout trainer / split), status workflow
Contact Messages	View all submissions with search, delete capability, email also sent to contact@airtrainr.com
Country Access	Toggle 15 countries with flags (US, CA, UK, AU, IN, DE, FR, BR, MX, JP, NZ, IE, ZA, AE, SG). Location Demand dashboard showing requests from blocked regions.
Platform Settings	Fee %, max booking distance, cancellation policy hours, dispute resolution days, support email, auto-approve toggle, maintenance mode, require verification toggle

04

SECTION FOUR

Business Workflows

End-to-end user journeys and system interactions

4.1 DIRECT BOOKING FLOW

1 Trainer sets availability (Recurring Weekly OR Per-Slot) 2 Athlete searches by sport/location 3 Views trainer profile with green calendar dates 4 Selects date, duration, sport, location 5 Stripe Checkout (price + 3% fee) 6 Webhook creates booking + payment_transaction (escrow) 7 Both receive email receipt 8 Session completed 9 Escrow released after hold period

4.2 TRAINING OFFER FLOW

1 Trainer browses athletes 2 Creates offer (sport, date/time, rate, message) 3 Athlete receives notification 4 Opens offer modal (sees sessions, trainer info, rate) 5 Clicks "Accept" → redirected to Stripe Checkout 6 Payment confirmed → webhook creates booking + updates offer status 7 Both receive email receipts + notification

4.3 CAMP ENROLLMENT FLOW

1 Trainer creates camp (name, per-day schedule, location, price, max spots) 2 Sends camp offer — dates/times auto-populated from camp schedule 3 Athlete sees read-only schedule preview 4 Accepts → Stripe payment

5 Webhook: fresh DB read → check spotsRemaining → if > 0, decrement + create booking → if = 0, auto-decline offer 6 Retry loop (3x) handles concurrent acceptances

> Race Condition Protection: When two athletes accept the last camp spot simultaneously, the retry loop ensures only one succeeds. Each attempt freshly reads spotsRemaining from the database, checks if > 0, then decrements. If the write fails (stale data), it retries with a fresh read. After 3 failures, the operation is rejected.

4.4 LOCATION INTELLIGENCE FLOW

1 User types city name 2 Frontend calls /api/places/autocomplete (server proxy) 3 Google Places returns predictions (250ms debounce) 4 User selects city 5 /api/places/details returns full address (city, state, country, ZIP, lat/lng) 6 If country not in allowed_countries → “Not available in your area” banner + “Notify me” lead capture 7 ZIP auto-filled + cross-validated against city country 8 Travel radius auto-switches to km for Canadian postal codes

05

SECTION FIVE

Payment Architecture

Financial model, escrow system, and Stripe integration

5.1 Fee Model

Party	Pays	Receives	Example ($50/hr session)
Trainer	$0	100% of session price	Receives $50.00
Athlete	Session + 3% fee	Training session	Pays $51.50
AirTrainr	$0	3% platform fee	Keeps $1.50

> Note: The 3% fee is configurable by the admin in Platform Settings. Trainer always receives their full rate — the platform fee is charged on top to the athlete.

5.2 ESCROW & HOLD SYSTEM

Trainer Type	Hold Period	Release Trigger
New trainer (< 3 sessions)	72 hours after payment	Session completed + hold expired
Established (3+ sessions)	24 hours after payment	Session completed + hold expired
Admin override	Immediate	Manual release from admin panel

5.3 STRIPE CONNECT ONBOARDING

Trainers connect their bank account through Stripe Express — a hosted onboarding flow managed entirely by Stripe. AirTrainr never sees, stores, or processes bank account details. The Payment Settings page shows three states:

NOT CONNECTED

"Connect Bank Account" button → creates Express account via /api/stripe/connect → redirects to Stripe hosted onboarding

PARTIALLY CONNECTED

Amber warning → "Complete Setup" button → resumes Stripe onboarding where left off

FULLY CONNECTED

Green "Payouts Enabled" status → bank last4 + name displayed → Stripe dashboard link

EMAIL RECEIPTS

After every payment, both athlete and trainer receive HTML email receipts via GoDaddy SMTP with full cost breakdown, session details, and trainer payout amount.

06

SECTION SIX

Location Intelligence

Geolocation, country management, and internationalization

6.1 Google Places Integration

All location services are proxied through server-side API routes (/api/places/*) to keep the Google API key secure and avoid CORS issues. The LocationAutocomplete component provides a unified interface across all pages — registration, profile editing, trainer setup, and camp location.

Feature	Implementation
City Autocomplete	Server proxy → Google Places Autocomplete → 5 predictions, 250ms debounce
Place Details	On selection → fetches city, state, country, ZIP, lat, lng from address_components
Country Filtering	Fetches allowed_countries from platform_settings → blocks results from non-allowed countries
Lead Capture	Blocked users see “Not available” banner → “Notify me” button → saves to location_leads table
ZIP Validation	Cross-validates postal code format (US/CA regex) against city country from Google API
Radius Units	Auto-detects Canadian postal codes → switches display to km. Stores internally as miles.

6.2 COUNTRY ACCESS MANAGEMENT

Administrators can enable/disable countries from the Settings page. 15 countries are preconfigured with flags: US, CA, GB, AU, IN, DE, FR, BR, MX, JP, NZ, IE, ZA, AE, SG. The Location Demand dashboard aggregates leads from blocked regions, sorted by request count, helping inform expansion decisions.

07

SECTION SEVEN

Security & Compliance

7.1 Security Architecture

Layer	Measure	Details
Database	Row Level Security (RLS)	All 15+ tables have Supabase RLS policies. Users can only read/write their own data.
Authentication	Password + OAuth	12+ char with uppercase, lowercase, number, special char. Google OAuth as alternative. No Apple login.
API Security	Role verification	All payment/connect APIs verify user exists AND has correct role (trainer) before processing.
Payment Security	Stripe Express	Bank details entered on Stripe hosted page. AirTrainr never sees/stores bank information. No Stripe account IDs exposed to frontend.
API Key Protection	Server-side proxy	Google Places API key never sent to browser. All calls proxied through /api/places/* routes.
Service Role	Server-only	Supabase service_role key used only in API routes + webhooks. Anon key for client with restricted RLS access.
Input Validation	Multi-layer	Client-side (form validation) + server-side (API checks) + database constraints (CHECK, NOT NULL, foreign keys)
Webhook Security	Signature verification	Stripe webhook events verified with STRIPE_WEBHOOK_SECRET before processing.

7.2 FORM VALIDATION SUMMARY

NAME FIELDS

Required, minimum 2 characters, inline red error text

PHONE

Optional, 10+ digits pattern match, inline error

ZIP/POSTAL CODE

US (5-digit) or CA (A1A 1A1) format. Red border + error if invalid. Cross-validated against city country.

PASSWORD

12+ chars, uppercase + lowercase + number + special character required

SECTION EIGHT

Testing & Deployment

QA process, test coverage, and infrastructure

8.1 E2E Testing (Playwright)

47 Playwright E2E tests organized in 6 test suites covering the complete application:

Suite	Tests	Coverage
01-auth	5	Login page, register (no Apple), athlete/trainer/admin login
02-athlete-profile	11	Dashboard, profile fields, avatar, autocomplete, sports, bookings, payments, contact, sub-accounts, messages
03-trainer-profile	10	Setup, ZIP validation, camps, availability, offers, subscription, earnings, payment settings, contact
04-offer-flow	7	Browse athletes, offer modal, date/time (no preferred slots), camp auto-fill, timezone, Stripe redirect
05-admin	10	Dashboard, trainers, modal z-index, athletes location, payments, sports, contacts, settings, map
06-ui-consistency	4	No scrollbars, italic headings, sport capitalization, Founding 50 priority

> Results: 34/47 tests passing. 13 admin tests fail due to test credential configuration (admin role not set in test DB). All athlete + trainer features verified.

8.2 DEPLOYMENT INFRASTRUCTURE

Service	Provider	Configuration
Web Hosting	Vercel	Auto-deploy from main branch, edge network, serverless functions
Database	Supabase	Hosted PostgreSQL 15, PostGIS, Realtime enabled, 500MB free tier
Payments	Stripe	Connect Express + Checkout, test mode + production keys
Media CDN	Cloudinary	Cloud: dslgstzex, Preset: airtrainr, Auto-format + quality optimization
Email	GoDaddy SMTP	smtpout.secureserver.net:465, SSL, contact@airtrainr.com
Domain	airtrainr.com	Production website + API endpoints

8.3 ENVIRONMENT VARIABLES

All secrets managed via Vercel Environment Variables (Production + Development). 14 env vars configured: Supabase URL/keys, Stripe keys/webhook secret, Cloudinary config, Google Places key, SMTP credentials, App URL.

DIGITAL MARKETING HEROES

"We Launch Brands in New Orbits"

digitalheroes.co.in

© 2026 Digital Marketing Heroes · All Rights Reserved · Made by Kayum Khan Sayal